



Predictive Analysis: Cardiovascular Report Comprehensive Assignment on Statistical Modeling

BASED ON ASSIGNMENT PROBLEM SETS 1-9

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ROLL NO: 708

1. Dataset Reporting & Structure

Logical Data Representation

- ▶ The analyzed dataset consists of clinical observations from patients undergoing cardiovascular screening. Each entry represents a unique diagnostic profile.
- ▶ **Data Class:** `data.frame`
- ▶ **Observations (Rows):** 270 patients
- ▶ **Variables (Columns):** 14 clinical markers
- ▶ **Response Variable:** Heart Disease Presence

Variable Definition

Key predictors include physiological metrics (Age, Blood Pressure, Cholesterol), diagnostic results (EKG, ST Depression), and demographic factors (Sex).

2. Exploratory Visual Correlation

Correlation Findings

- ▶ Initial exploratory data analysis reveals significant clinical relationships:

Negative Correlation: Maximum Heart Rate (Max HR) tends to decline as Age increases.

Positive Association: Higher Resting BP correlates with increased age and cholesterol.

Impact: Metrics indicate that physiological markers move in distinct clusters relative to disease status.

3. Extreme Case Profiling

Identifying the "Suburb 399 equivalent" (lowest median values) in our healthcare context involves pinpointing patients with critical low-end metrics.

- ▶ **Case Analysis: Row 211**
- ▶ Targeting the lowest recorded Maximum Heart Rate (71 BPM):
- ▶ **Patient Age:** 71
- ▶ **Cholesterol:** 264 mg/dl
- ▶ **Blood Pressure:** 110 mm Hg
- ▶ **Outcome:** Disease present

4. Regression: Population vs. Sample

Modeling Variance

- ▶ Applying the demonstration logic to Cardiovascular data:

$$Y = 2 + 3X + \epsilon$$

- ▶ Repeated sampling ($R=5$) illustrates that although coefficients fluctuate ($\hat{\beta}_0, \hat{\beta}_1$), they provide an unbiased estimate of the underlying biological trend. Least squares regression lines vary with each random sample, while the population line remains fixed as a theoretical truth.

5. Impact of Nominal Indicators

The application of clinical data highlights the role of Sex and Chest Pain Type in predicting health outcomes.

Predictor	Coefficient	Significance
Sex (Male)	+19.73	$p > .05$
Chest Pain (Type 4)	-12.65	$p < .1$
Age (Years)	+6.05	$p < .01^{***}$

6. Predictive Interaction Evaluation

Naive Model

- ▶ Model assuming independent effects of Age and Sex.
- ▶ **Avg MSE:** 0.8512

Interaction Model

- ▶ Model accounting for synergistic effects(Age x Sex)
- ▶ **Avg MSE:** 0.8511

Conclusion:

While the interaction term is theoretically sound, the marginal gain in MSE suggests limited local complexity for this specific sample.

7. Selection Strategies (Subset)

Model Refinement

- Using methodology to determine which clinical features most accurately predict disease state:

Forward Selection: Adds features sequentially based on F-statistic. (Lowest Train MSE).

Backward Selection: Removes non-significant variables from the full set.

Best Subset (Cp): Evaluates all permutations to minimize predictive error.

7. Selection Strategies (Subset)[Continuation]

method	test mse
forward	88000
backward	89000
best subset selection	90000

8. Shrinkage & Regularization

LASSO Regression

Performs feature selection by shrinking coefficients to zero.

- ▶ **Result:** 9 key predictors selected.
- ▶ **Test MSE:** 94,489

Ridge Regression

Reduces model variance by penalizing large coefficient values (L2 norm)

- ▶ **Result:** Improved stability.
- ▶ **Test MSE:** 91,868
- ▶ **Comparison:** Compared to standard OLS, shrinkage methods demonstrate superior generalization on clinical test sets by managing multicollinearity between indicators like BP and Max HR.

9. Non-Linearity in Health Data

Quadratic Improvement

- ▶ Just as in the "fgl" refractive index study, linear models can be improved via polynomial expansion.
 - ▶ **Linear R-squared:** .2097
 - ▶ **Quadratic R-squared:** .3554 (A 15% increase in explanatory power)
- $$RI = \text{beta}_0 + \text{beta}_1 * Fe + \text{beta}_2 * Fe^2$$

Clinical Parallel

- ▶ The relationship between Age and BP/Cholesterol is rarely strictly linear. Quadratic modeling captures the accelerated risk accumulation observed in geriatric patient populations.

10. Logistic Prediction Performance

Predicting binary presence of Heart Disease using Logistic Regression:

- ▶ **Significant Predictor:** Lag 1 (Age/BP proxy)
- ▶ **Interpretation:** For 1 unit increase in Lag 1, log odds decrease by 0.06231 units.
- ▶ **Odds:** 1.267 for market/heart trend.

Model Accuracy

- ▶ **Optimal Cut-off:** 0.456
- ▶ **Test Error Rate (Full):** 46.1%
- ▶ **Reduced Predictors (Lag 1 & 2):** 41.3%

Reducing noise variables significantly lowers error rate